

Assignment 11:

1. Set up Firebase authentication in a new React app and use `onAuthStateChanged()` inside a `useEffect` to detect if a user is logged in or not; display 'Welcome, Guest' or 'Welcome, [User Email]' accordingly in the navbar.
2. Implement a Logout button in the navbar that signs out the user using Firebase's `signOut()` method and updates the UI to reflect the logged-out state.
3. Build a Spotify-style greeting: after login, show the user's display name or email in the navbar, and after logout, instantly switch to 'Sign In' without needing a page refresh.

***Hint:** Use `onAuthStateChanged()` to listen for authentication state changes and update your React state accordingly.*
4. Secure your app by ensuring that any protected page (like `/profile` or `/myorders`) automatically redirects to the login page if the user is not authenticated, using the current authentication state from `onAuthStateChanged()`.

Answer:-

1, 2, 3.-

```
import { onAuthStateChanged, signOut } from 'firebase/auth';
import { useEffect, useState } from 'react';

const FireBaseExercise = () => {
  const [user, setUser] = useState(null);
  useEffect(() => {
    const unsubscribe = onAuthStateChanged(auth, (currentUser) => {
      setUser(currentUser);
    })
    return () => unsubscribe();
  }, [])

  const handleSignOut = async () => {
    try {
      await signOut(auth);
      console.log("Logout succesFully...");
    } catch (error) {
      console.log(error.message);
    }
  }

  return (
    <>

    <nav>
```

```

        {
          user ? `Welcome ${user.email}` : "welcome Guest"
        }
      </nav>
      {
        user && (
          <button onClick={handleSignOut}>SignOut</button>
        )
      }

      {
        user ? `welcome ${user.displayName || user.email}` : `Sign In`
      }
    </>
  )
}

export default FireBaseExercise;

```

4. protected Routing

Protected.jsx

```

import { Navigate } from 'react-router-dom';

const Protected = ({user, children}) => {
  if(!user){
    return <Navigate to="/login"/>
  }
  return children
};

export default Protected

```

Head.jsx

```

import { useEffect, useState } from "react";
import {
  BrowserRouter,
  Routes,
  Route,
} from "react-router-dom";

import { auth } from "../firebase";
import { onAuthStateChanged } from "firebase/auth";

```

```

import Login from "./Login";
import Profile from "./Profile";
import MyOrders from "./MyOrders";
import ProtectedRoute from "./ProtectedRoute";

function App() {
  const [user, setUser] = useState(null);

  useEffect(() => {
    const unsubscribe = onAuthStateChanged(
      auth,
      (currentUser) => {
        setUser(currentUser);
      }
    );

    return () => unsubscribe();
  }, []);

  return (
    <BrowserRouter>
      <Routes>
        <Route
          path="/login"
          element={<Login />}
        />

        <Route
          path="/profile"
          element={
            <ProtectedRoute user={user}>
              <Profile />
            </ProtectedRoute>
          }
        />

        <Route
          path="/myorders"
          element={
            <ProtectedRoute user={user}>
              <MyOrders />
            </ProtectedRoute>
          }
        />
      </Routes>
    </BrowserRouter>
  );
}

```

```
}
```

```
export default App;
```